

E-Commerce Logistics Performance Dashboard

Delivery Operations Analysis Using Power BI

Field	Details
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Program	PGDM - Operations & Analytics, SJIM Bangalore
Tool	Microsoft Power BI Desktop
Dataset	E-Commerce Shipping Data, Kaggle (prachi13/customer-analytics)
Dataset Size	10,999 rows 12 columns
Date	September 2025

1. Project Overview

This project analyses 10,999 shipment records from an international e-commerce electronics company to measure delivery performance, identify operational inefficiencies, and generate data-backed recommendations. The output is a two-page interactive Power BI dashboard designed for both executive review and root cause analysis.

Dimension	Detail
Business Problem	High delivery delay rate with no clear operational visibility into root causes
Objective	Identify which warehouses, shipment modes, and product segments are driving delays
Output	Two-page Power BI dashboard - Executive Summary + Analysis
Approach	Data cleaning in Power Query → DAX measures → KPI visualisation → Recommendations

2. Dataset

2.1 Source

Kaggle - E-Commerce Shipping Data by prachi13. Publicly available.

URL: [kaggle.com/datasets/prachi13/customer-analytics](https://www.kaggle.com/datasets/prachi13/customer-analytics)

2.2 Column Reference

Column	Type	Description
Warehouse_block	Categorical	Warehouse identifier (A, B, C, D, F)
Mode_of_Shipment	Categorical	Transport mode - Ship, Flight, Road
Customer_care_calls	Numeric	Number of support calls per order - escalation proxy
Customer_rating	Numeric (1–5)	Post-delivery satisfaction rating

Column	Type	Description
Cost_of_the_product	Numeric	Product price in USD
Prior_purchases	Numeric	Number of previous orders by same customer
Product_importance	Categorical	Company classification - Low / Medium / High
Gender	Categorical	Customer gender
Discount_offered	Numeric	Discount percentage applied to order
Weight_in_gms	Numeric	Shipment weight in grams
Reached.on.Time_Y.N	Binary (0/1)	Core outcome variable, 1 = On Time, 0 = Delayed

2.3 Data Cleaning (Power Query)

- Renamed 'Reached.on.Time_Y.N' to 'On_Time' to avoid DAX formula errors
- Created 'Delivery_Status' column - converted 0/1 to 'Delayed' / 'On Time' for visual readability
- Created 'Weight_Category' - grouped shipments into Light (<2kg), Medium (2–4kg), Heavy (>4kg)
- Created 'Discount_Range' - grouped discount % into Low (0–10%), Medium (11–30%), High (31–50%), Very High (>50%)
- Verified all column data types - numeric columns set to Whole Number, categorical to Text

3. DAX Measures

Measure	Formula	Purpose
Total Orders	COUNTROWS('Train ')	Total shipment count
On Time Orders	COUNTROWS(FILTER('Train', 'Train'[On_Time] = 1))	Count of on-time deliveries
Delayed Orders	COUNTROWS(FILTER('Train', 'Train'[On_Time] = 0))	Count of delayed deliveries

On Time Rate %	DIVIDE([On Time Orders], [Total Orders]) * 100	Core KPI - overall delivery performance
Delay Rate %	100 - [On Time Rate %]	Inverse of on-time rate
Avg Customer Rating	AVERAGE('Train'[Customer_rating])	Mean satisfaction score
Avg Care Calls	AVERAGE('Train'[Customer_care_calls])	Mean escalation calls per order

4. Dashboard Structure

4.1 Page 1 - Executive Summary

Visual	Type	Fields Used	Business Question
KPI Cards (×4)	Card	Total Orders, On Time Rate %, Delay Rate %, Avg Rating	Overall operation health at a glance
On-Time Rate by Warehouse	Clustered Bar	Warehouse_block, On Time Rate %	Which warehouse is underperforming?
Delay Rate by Product Importance	Clustered Bar	Product_importance, Delay Rate %	Are high priority orders protected?
Shipment Mode Split	Donut	Mode_of_Shipment, Total Orders	How are orders distributed by mode?
Warehouse Slicer	Slicer (Tile)	Warehouse_block	Filter all visuals by warehouse
Shipment Mode Slicer	Slicer (Dropdown)	Mode_of_Shipment	Filter all visuals by transport mode

4.2 Page 2 - Deep Dive Analysis

Visual	Type	Fields Used	Business Question
Customer Rating vs Delivery Status	Clustered Column	Customer_rating, Total Orders, Delivery_Status	Do delays lower customer ratings?
Escalation Calls by Warehouse	Clustered Bar	Warehouse_block, Avg Care Calls	Which warehouse generates most complaints?
Delay Rate by Weight Category	Clustered Bar	Weight_Category, Delay Rate %	Do heavier shipments delay more?
Warehouse: Delay vs Satisfaction	Scatter	Avg Customer Rating (X), Delay Rate % (Y), Warehouse_block (Values)	Do high-delay warehouses also have low ratings?

5. Key Findings

Finding 1 - Overall Delay Rate: 59.7%

10,999 orders analysed. Only 40.3% reached customers on time. A delay rate approaching 60% indicates a structural operational failure, not isolated incidents.

Finding 2 - Warehouse A is the Weakest Performer

Warehouse A recorded the lowest on-time rate (~57%) and the highest average customer escalation calls across all warehouse blocks. It is the clearest operational priority.

Finding 3 - Priority System is non-functional

High importance products show a delay rate virtually identical to Low importance products. The company's product importance classification has zero measurable impact on delivery performance.

Finding 4 - Customer Satisfaction is multi-factorial

Average ratings vary narrowly between 2.96 and 3.01 across warehouses despite significant delay rate differences. Delivery timeliness is one driver of satisfaction, not the only one.

Finding 5 - Ship Mode Carries Highest Delay Risk

Ship mode shows the highest delay rate relative to Flight. The cost-speed tradeoff between modes is not being managed against product value or customer SLA requirements.

Finding 6 - Weight Has Marginal Impact on Delays

Heavy shipments (>4kg) show marginally higher delay rates, but the difference is not significant. Delays are process-driven, not product-weight driven.

6. Recommendations

Recommendation	Priority	Expected Impact
Conduct process audit of Warehouse A - review pick-pack cycle times, dispatch cutoff adherence, and carrier handover protocols	High	Reduce Warehouse A delay rate by 8–12%
Implement a dedicated fast-track dispatch lane for High importance products with committed same-day dispatch SLA	High	Eliminate priority gap - align operational output with classification intent
Introduce shipment mode segmentation - products above a value threshold default to Flight unless customer selects standard	Medium	Reduce delay rate on high-value orders without blanket cost increase
Build a composite warehouse scorecard tracking on-time rate + escalation rate + customer rating weekly	Medium	Replace single-metric monitoring with multi-dimensional performance visibility

7. Limitations & Scope for Further Analysis

- **No Timestamps:** Dataset contains no order creation, dispatch, or delivery timestamps. Actual lead time analysis is not possible - only binary on-time/delayed classification.

- **No Geography:** No city or region data. Tier-2/Tier-3 city performance differences cannot be analysed - a critical gap for Indian e-commerce context.
- **No Returns Data:** On-time delivery without returns data gives an incomplete picture of logistics performance. High return rate categories (e.g. fashion) are not captured.
- **No Trend Data:** Single snapshot - no time series. Seasonal performance patterns (sale events, peak periods) cannot be identified.

8. Project Files

File	Description
Train.csv	Raw dataset - sourced from Kaggle (prachi13/customer-analytics)
Logistics_Performance_Dashboard.pbix	Power BI dashboard file - two pages, 8 visuals, 7 DAX measures
Logistics_Dashboard_Report.docx	This report - methodology, findings, recommendations

Tools: Microsoft Power BI Desktop | Data Source: Kaggle (Public) |